



DUNMAN HIGH SCHOOL
Preliminary Exam
Year 6

H2 Biology

9744/04

Paper 4 Practical

Answer Scheme

1 (a)

Sketch

- Axes: y-axis labelled 'rate of photosynthesis' AND x-axis labelled 'light intensity'
- Line: line starts at **origin (0, 0)** with positive gradient and then levels off or levels off then dips

Explanation

- (at lower light intensities) as light intensity increases, there is more photolysis / reduction of NADP; OWTTE.
- (at higher light intensities) light is no longer a limiting factor / some other factor limits the rate of photosynthesis / reference to photobleaching

1 (b)

- Correct heading for independent variable (percentage light intensity / %) in first column
Correct heading for dependent variable (distance in mm or cm) in next column
- Records five readings
- Records distance (consistently within column) to nearest mm or 0.5 mm
- (*Trend*) Records greatest distance for filter F1 AND shortest distance for filter F5

1 (c)(i)

1. Axes: Both axes correctly labelled including units
e.g. y-axis labelled 'Distance travelled by meniscus / mm' AND x-axis labelled 'percentage light intensity / %'
2. Scale: Axes scaled appropriately so that the graph occupies at least 50% of the grid and divisions are equidistant
3. Plot: All data points correctly plotted
4. Line: Appropriate line of best fit (curve);

1 (c)(ii)

Consider candidates' answers to 1(a) and allow:

- Appropriate specific comment on match of results to the proposed relationship;
- Appropriate comment on how this affects confidence in the hypothesis / relationship;

E.g.

Howard's results show the same pattern as that predicted in 1(a) which increases confidence in the hypothesis / relationship.

1 (d)

Limiting factor is light intensity (no mark on its own)

Since as light intensity increases, the rate of photosynthesis increases; OWTTE

1 (e)

Any 2:

- Sodium hydrogencarbonate breaks down to release CO₂
- So CO₂ is not / less likely to be limiting
- Idea that light intensity will be the only variable

1 (f)

Any 3:

- Repeat with control syringe lacking pondweed to check that the results are due to (photosynthesis of) the pondweed;
- Repeat with control syringe lacking pondweed AT THE SAME TIME, to allow for movement of meniscus due to changes in air pressure / temperature;
- Eliminate all other light sources / ensure all other light sources are constant, so that the light intensity is as expected;
- Use fresh hydrogencarbonate solution each time to ensure that (starting) CO₂ concentration is constant;

2 (a)

Similarities (max 1):

- Air spaces are present in both specimens
- Air spaces in both specimens are randomly distributed
- Both specimens lack vascular bundles

Differences (max 1):

- There are more air spaces in the specimen on slide S1 than the specimen in Fig. 2.1
- The stem of the specimen in Fig. 2.1 is circular in shape while the stem of the specimen on slide S1 is quadrangular (OWTTE) in shape

2 (b)(i)

Diameter = $W = 7 \text{ cm}$

Correct measurement of diameter using W line;

Radius, $r = W/2 = 3.5 \text{ cm}$

Actual radius of stem, $R = r/15$

Correct working / formula;

= **0.23 cm**

Display final result with correct d.p. and units;

2 (b)(ii)

Diameter of field of view = **2.0 mm**; (Accept! 1.9 – 2.1 mm)

2 (b)(iii)

Fraction of diameter of field of view = **2/5**; (Accept! fraction or decimal, Range between 0.4 to 0.45)

2 (b)(iv)

Radius of S1 = $2/5 \times 2.0 \text{ mm} = \textbf{0.8 mm}; (Accept! 1 or 2 d.p.)$

2 (b)(v)

Any 3:

Additional apparatus: E.g. Eyepiece graticule, Stage micrometer

- Explanation of how to calibrate eyepiece graticule
- Explanation of how to measure specimen using graticule and relevant calculation
- Record measurement for several different radii and calculation of mean

2 (c)(i)

1. Line (drawings must be drawn with continuous line);
2. Space (maximal use of space horizontally / vertically);
3. Proportion (e.g. distribution of tissues must be proportionate and consistent);
4. Accuracy (only three air spaces and arrangement of tissues)

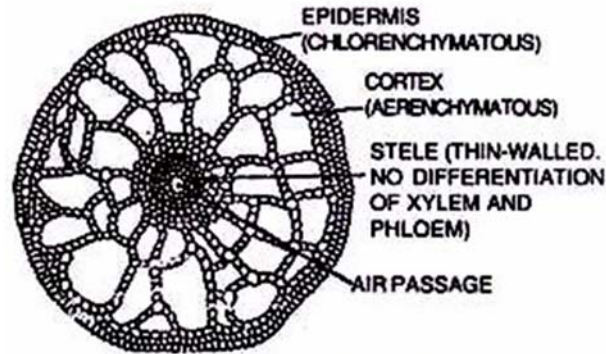


Fig. 2.4. T.S., of *Elodea* stem. (Note undifferentiated vascular tissue in the stelar region)

2 (c)(ii)

1. Line (drawings must be drawn with continuous line);
2. Space (maximal use of space horizontally / vertically);
3. Proportion (e.g. cell wall must be proportionate and consistent);
4. Accuracy (three touching cells, cell wall)

2 (d)

Assist in the exchange of gases for photosynthesis;

Used for buoyancy;

Max 1

3. Planning

Award 1 mark for each of the following (max 14):

1. States that independent variable is light intensity AND uses at least five different light intensities uniformly spaced or derived;
e.g. regularly increasing distance from light source
2. Discusses ways of measuring light intensity;
e.g. use of light meter / calibration curve (intensity vs distance) / mathematical relationship between distance to lamp and light intensity
3. Colour of hydrogencarbonate indicator solution;
4. Identifies at least TWO variables to control;
e.g. area of leaves, wavelength of light, duration of light, volume of indicator, temperature
5. Describes how variables (identified in 4) are controlled;
6. Refers to control experiment without leaves;
7. Plans suitable method to vary light intensity;
8. Devises a method that uses tubes sealed with bungs (or equivalent) containing leaves and hydrogencarbonate indicator solution;
Accept! any relevant diagram
9. Plans a suitable procedure that involves monitoring the colour of indicator solution in response to different light intensities over a set periods of time (5-10min) or at set time intervals;
10. Specifies method of monitoring colour of indicator solution;
e.g. use of colour standard
11. States that starting colour of hydrogencarbonate indicator solution MUST be red;
12. Compensation point will be light intensity at which there is no change in the colour of hydrogencarbonate indicator solution over (entire) period monitored;
13. Reference to repeating at least two more times with different leaves;
14. Shows how results are to be presented in the form of a table with independent (light intensity) and dependent variables (colour change) in appropriate columns;
15. Repeat using narrower range of light intensities after identifying broad range within which the compensation point lies;
16. Refers to hazard AND precaution;
e.g. heat from bulbs / electricity and water